

browser, except for security updates (of which quite a lot were required). Microsoft belatedly tried to get back into the game. If a user tried to change the browser to Chrome, some builds of Windows 10 would throw a pop-up asking the user to 'Give Microsoft Edge a shot'. On Twitter, I have seen photos of the Bing search engine doing the same when someone tries to search for Chrome. "Learn the facts!" Bing would thunder.

The origins of the Web

The National Centre for Supercomputing Applications (NCSA) at the University of Illinois created the Mosaic browser in 1993. Marc Andreessen, who led the development of Mosaic, left NCSA and started work on the free but closed source Netscape browser based on Mosaic (a proprietary browser, but its source code was publicly available for non-commercial use). Netscape was released in 1994 and became the dominant browser by 1995, eclipsing Mosaic. In the same year, Microsoft released Internet Explorer, based on a Mosaic licence.

The Web initially served static Web pages. Later, CGI programs were used to serve dynamically generated Web pages. These CGI programs could not handle a lot of traffic and more sophisticated server-side scripting languages were developed. Similarly, for the browser, JavaScript was used as a client-side scripting language.

JavaScript was developed by Brendan Eich while working at Netscape. He released the first version in a matter of 10 days. At that time, Java had captured the imagination of technologists with its 'Write Once, Run Anywhere' promise. Java applets could be downloaded from the Web and run on any computer with a Java runtime. Java applets could also run inside Web browsers. Netscape decided to ride on Java's phenomenal name-recognition and named its native client-side scripting technology as JavaScript. This creates some confusion even now. Eventually, JavaScript became the native scripting technology for browsers. DHTML (HTML+JavaScript) was touted as 'HTML with steroids'.

Microsoft copied JavaScript's API and created JScript for use with IE and other Microsoft products. Thus, IE supported JavaScript/JScript nomenclature in addition to VBScript. In 1996, Microsoft also introduced a proprietary technology called ActiveX in IE 3.0. It enabled Windows-only software components (COM-based DLLs and OCX files) to be used within Web pages. This meant that Web pages that used ActiveX components would not run on non-Windows operating systems. Even on Windows, browsers such as Firefox and Opera failed to display such Web pages. Everyone had to have IE and Windows. However, ActiveX introduced a host of security problems. Malware writers used IE to infect Windows systems.

Firefox: The past

In its heydays, AOL acquired several software products



Figure 1: Google page asks the user to download Chrome



Figure 2: Microsoft gives a shot at protecting its Edge

including Netscape, ICQ and Winamp. By 2003, AOL's fortunes had waned and it created the Mozilla Foundation to ensure that development of Mozilla Firefox would independently continue as an open source browser project. Since then, Mozilla has been

on a mission to 'take back the Web'. Many Netscape/Firefox and open source fans adopted the browser and helped it spread. In 2004, Mozilla agreed to make Google its default search engine and this earned it millions of dollars in revenue. Later, Mozilla diversified this revenue stream by signing up with the Yahoo, Yandex and Baidu search engines. The Netscape Communicator mail client was rebranded as Thunderbird. Its main focus was to become a free alternative to Microsoft Outlook.

Meanwhile, the Conqueror Desktop Environment (KDE) project decided to write its own browser engine. It was called KHTML, and was used by the Konqueror browser and file manager. Apple forked the open source KHTML and created WebKit. From then on, WebKit became a complicated beast with numerous Mac-related dependencies.

When Google decided to create the Chrome browser, it adopted the WebKit rendering engine. For some time, Google worked with Apple and other WebKit developers. In April 2013, Google announced that it was forking WebKit and creating the Blink rendering engine. Opera and other Chrome clones followed suit.

Before becoming a Chrome clone, Opera was a pioneering, if not popular, Web browser. It had an extremely customisable user interface that fans loved. Opera had its own proprietary rendering engine called Presto. It was not just a browser. Opera bundled an e-mail